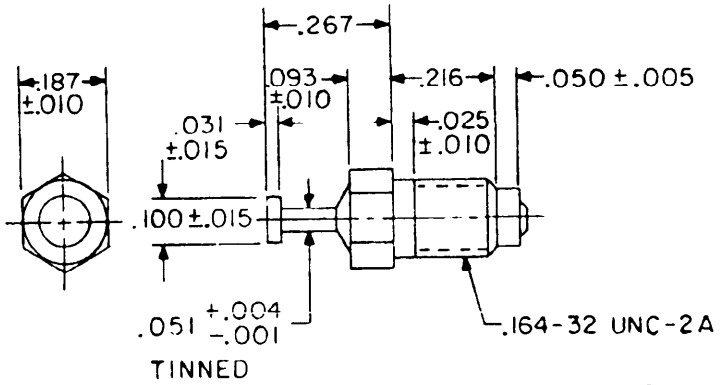
MILITARY SPECIFICATION SHEET

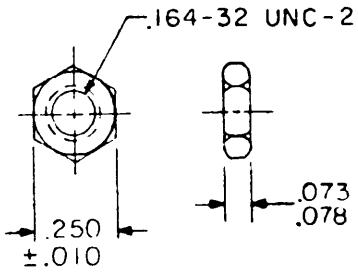
CAPACITORS, FIXED, CERAMIC DIELECTRIC, (GENERAL PURPOSE), ESTABLISHED RELIABILITY STYLE CKR83 (STANDOFF)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

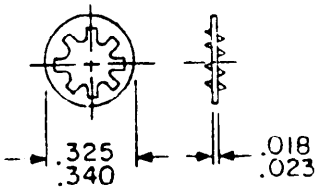
The complete requirements for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-C-39014.



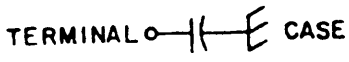
INCHES	MM	INCHES	MM
.001	.03	.073	1.85
.004	.10	.078	1.98
.005	.13	.093	2.36
.010	.25	.100	2.54
.015	.38	.164	4.17
.018	.46	.187	4.75
.023	.58	.216	5.49
.025	.64	.250	6.35
.031	.79	.267	6.78
.050	1.27	.325	8.26
.051	1.30	.340	8.64



HEX NUT



LOCKWASHER
MS 35333-72



CIRCUIT DIAGRAM



APPROX ACTUAL SIZE

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ±.031(.79 mm).
3. A nut and internal tooth lock washer shall be supplied with each capacitor.

FIGURE 1. CKR83 capacitors.

(A) denotes changes

FSC 5910

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case type: Standoff.

Capacitance value: See table I.

Capacitance tolerance: +80, -20 percent.

Operating temperature range: -55°C to +125°C.

Voltage rating: 200 volts dc at 85°C, derated linearly to 100 volts dc at 125°C (see figure 3).

Current rating: 10 amperes, maximum.

Dissipation factor: Not more than 3.0 percent.

Dielectric withstanding voltage:

Application of test voltage: Between one capacitor-element terminal and the case.

Barometric pressure:

Method 105, MIL-STD-202, condition B (50,000 feet).
Application of test voltage: Between one terminal and the case.
Test potential: 150 percent of dc rated voltage.

Insulation resistance:

Measured between one terminal and the case: Not less than the value specified in table I.

Insertion loss: Not applicable.

Seal: Not applicable.

Solderability: Method 208, MIL-STD-202, two terminals.

Voltage-temperature limits: In accordance with MIL-C-39014.

Capacitance change with reference to 25°C		For capacitance
Steps A through D of table V Bias = 0 volts	Steps E through G of table V Bias = rated voltage	
±15 percent	+15, -25 percent	Less than 1,000 pF
+22, -56 percent	+22, -66 percent	1,000 pF and greater

Immersion:

IR: Not less than 3,000 megohms.
ΔC: ±20 percent from initial measured value.

Salt spray (corrosion): In accordance with MIL-C-39014.

Terminal strength: Not applicable.

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Moisture resistance:

Application of test voltage: Between one terminal and the case.
IR: Not less than 3,000 megohms.
ΔC: ±20 percent from initial measured value.

Resistance to soldering heat:

ΔC: Less than ±20 percent from initial measured value.
Cooling time prior to measurement after test: 1 ±.1 hour.

Life:

Rated conditions: 100 percent of dc rated voltage at 85°C.

0 through 3,000 hours:

IR: At 25°C not less than 50,000 megohms.
At 125°C not less than 5,000 megohms.
ΔC: ±20 percent from initial measured value.
Dissipation factor: Not greater than 3.0 percent.

4,000 through 32,000 hours:

IR: At 25°C not less than 10,000 megohms.
At 125°C not less than 1,000 megohms.
ΔC: ±20 percent from initial measured value.
Dissipation factor: Not greater than 3.5 percent.

Accelerated conditions, 200 percent of dc rated voltage at 85°C.

0 and 250 hours:

IR: At 25°C not less than 50,000 megohms.
At 125°C not less than 5,000 megohms.
ΔC: ±20 percent from initial measured value.
Dissipation factor: Not greater than 3.0 percent.

1,000, 2,000 and 4,000 hours:

IR: At 25°C not less than 10,000 megohms.
At 125°C not less than 1,000 megohms.
ΔC: ±20 percent from initial measured value.
Dissipation factor: Not greater than 3.5 percent.

Marking: Marking shall consist of the "J" certification mark and the military part number as shown on figure 2.



FIGURE 2. Hex flats - expanded view.

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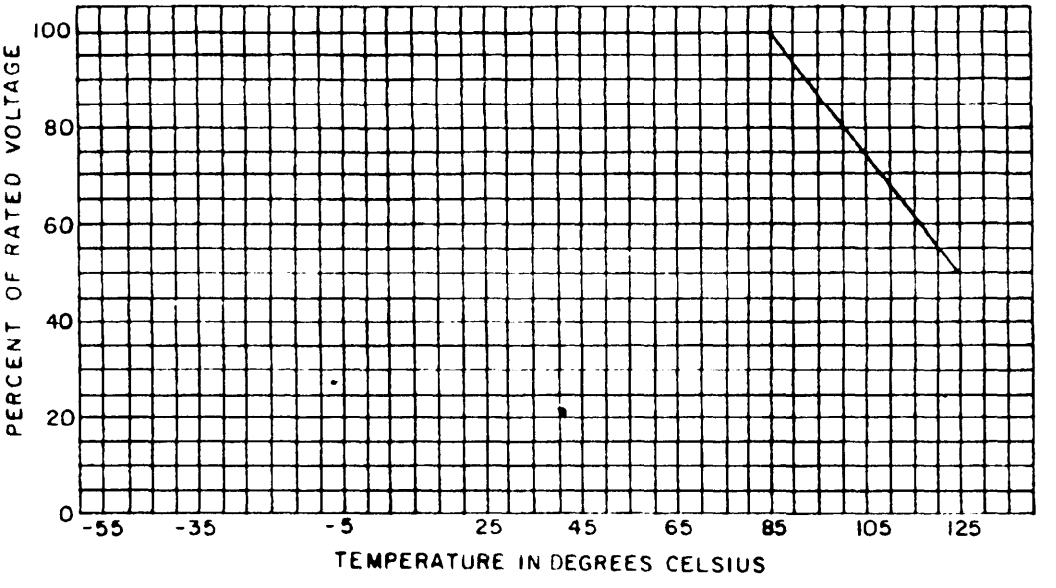


FIGURE 3. Voltage derating with temperature.

Part number: M39014/19- (dash number from table I).

NOTE: Installation torque should not exceed 5 inch-pounds.

TABLE I. Electrical characteristics and dash number.

Capacitance	Dash number			
	Failure rate level (percent per 1,000 hours)			
	1.0(M)	0.1(P)	0.01(R)	0.001(S)
pF				
10	0321	0341	0361	0381
22	0322	0342	0362	0382
47	0323	0343	0363	0383
100	0324	0344	0364	0384
220	0325	0345	0365	0385
470	0326	0346	0366	0386
1000	0327	0347	0367	0387
1500	0328	0348	0368	0388
5000	0329	0349	0369	0389
10000	0330	0350	0370	0390
15000	0331	0351	0371	0391
27000	0332	0352	0372	0392

Supersession data: Data numbers previously listed in this specification sheet were for capacitance values and tolerances that are not available for this style; no specific item to item replacements is recommended.

Revision letters are not used to denote changes due to the extensiveness of the changes.

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Custodians:
Army - ER
Navy - EC
Air Force - 11
NASA - NA

Review activities:
Navy - OS, TD
DLA - ES

User activities:
Navy - AS, SH, MC

Agent:
DLA - ES

Preparing activity:
Navy - EC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-39014/19A	CAPACITORS, FIXED, CERAMIC DIELECTRIC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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